

Thermodynamic properties of lattice hard-sphere models

A. Z. Panagiotopoulos^{a)}*Department of Chemical Engineering, Princeton University, Princeton, New Jersey 08540*

(Received 13 June 2005; accepted 30 June 2005; published online 12 September 2005)

Thermodynamic properties of several lattice hard-sphere models were obtained from grand canonical histogram- reweighting Monte Carlo simulations. Sphere centers occupy positions on a simple cubic lattice of unit spacing and exclude neighboring sites up to a distance σ . The nearest-neighbor exclusion model, $\sigma = \sqrt{2}$, was previously found to have a second-order transition. Models with integer values of $\sigma = 1$ or 2 do not have any transitions. Models with $\sigma = \sqrt{3}$ and $\sigma = 3$ have weak first-order fluid-solid transitions while those with $\sigma = 2\sqrt{2}$, $2\sqrt{3}$, and $3\sqrt{2}$ have strong fluid-solid transitions. Pressure, chemical potential, and density are reported for all models and compared to the results for the continuum, theoretical predictions, and prior simulations when available. © 2005 American Institute of Physics. [DOI: 10.1063/1.2008253]

I. INTRODUCTION

The one-component hard-sphere model is defined by infinite interparticle repulsion at distances less than the particle diameter σ and zero everywhere else. Despite the apparent simplicity and long history of the model, it continues to generate interest for simulations^{1,2} theory³ and experiments.⁴ In three dimensions, the model has a first-order fluid-solid phase transition⁵ with a relative density increase on crystallization of approximately 10%.

In recent years, several studies have examined the properties of “finely discretized” lattice models that interpolate between continuum and more commonly used “coarse” lattice models. The ratio of a characteristic particle size σ to a lattice parameter a is defined as $\zeta = \sigma/a$; the values of ζ significantly greater than 1 correspond to finely discretized lattices. Initial studies of ionic systems⁶ demonstrated that the location of the liquid-gas critical point is within a few percent of the continuum system for $\zeta \geq 3$, while qualitative differences were observed for the lower values of ζ . Subsequent theoretical^{7–9} and simulation studies^{10–12} have established the usefulness of such models for ionic systems. Finely discretized models have also been used to study simple atomic fluids,¹³ biomolecules,¹⁴ and polymers.¹⁵ The main attractive feature of such models is that the thermodynamic properties of continuum systems are closely reproduced at relatively low values of ζ , often with significant computational savings. Such savings result because of the reduction of the dimensionality of configuration space and the existence of specialized algorithms for evaluating intermolecular interactions for the lattice models.

The hard-sphere system has no direct interactions at distances greater than σ and is dominated by packing at short distances. One may reasonably expect that the presence of a lattice may lead to significant differences from the behavior of corresponding continuum systems, especially for ordered solid phases. The presence of an underlying lattice clearly

favors the crystalline phase and destabilizes the disordered fluid. Significant insights on the melting transition are obtained from the study of simple models.¹⁶

On a lattice, the $\zeta = 1$ “hard-sphere” model is trivially soluble and does not have any phase transitions. The “nearest-neighbor exclusion lattice gas” model ($\zeta = \sqrt{2}$) has been extensively studied in the past¹⁷ and continues to be the subject of theoretical interest.¹⁸ On the simple cubic lattice, the model has a second-order transition at a density $\rho\sigma^3 \approx 0.6$ —at higher density there is a coexistence between disordered and ordered sublattices. The universality class and critical exponents have been determined to be three-dimensional Ising by Heringa and Blöte.¹⁹ Models with exclusion effects beyond the first nearest neighbor have received much less attention. Hall and Stell²⁰ and Cowley²¹ have examined models with a variable exclusion range in two and three dimensions using mean-field methods. More recently, multisite exclusion hard-core models were studied using density-functional theories.^{22,23}

In the present work, several lattice hard-sphere models with discretization parameters ζ between $\sqrt{2}$ and $3\sqrt{2}$ were studied. The equation of state and phase transitions were obtained from grand canonical Monte Carlo simulations. The location and character of phase transitions between disordered and ordered phases were found to be quite sensitive to the value of the discretization parameter.

II. MODELS AND METHODS

The models studied were spheres of varying diameter, σ , on the simple cubic lattice of unit lattice spacing a , as shown in Fig. 1. With this choice of lattice spacing, the numerical values of ζ and σ are identical. The $\sigma = \sqrt{2}$ model has nearest-neighbor exclusion, while the $\sigma = \sqrt{3}$ model has next-nearest-neighbor exclusion. These correspond to a number of excluded sites $M = 7$ (1 central site + 6 nearest neighbors) and $M = 19$ (1 central site + 6 nearest neighbors + 12 next-nearest neighbors), respectively. Bigger spheres exclude more sites, all the way up to 171 sites for $\sigma = 2\sqrt{3}$, as shown in Table I.

^{a)}Electronic mail: azp@princeton.edu

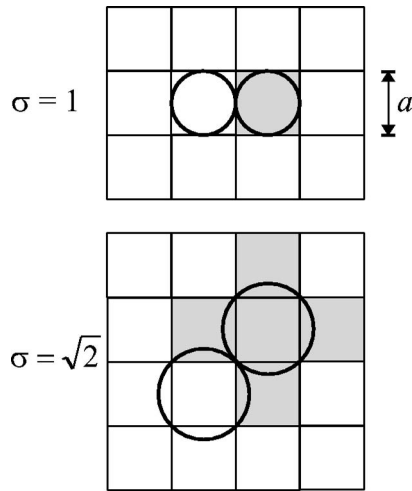


FIG. 1. Two-dimensional projection of two adjacent hard spheres for $\sigma=1$ and $\sigma=\sqrt{2}$. All systems studied were three dimensional. Shaded cells are excluded to centers of other spheres.

I used grand canonical Monte Carlo simulations combined with multihistogram reweighting, as described previously.²⁴ In brief, the simulations were performed in cubic boxes of volume $V=L \times L \times L$ under periodic boundary conditions. The lattice spacing, a , was taken as the unit of length. Since the systems are athermal, the temperature $\beta=1/kT$ does not enter as an independent variable, but only in combination with the chemical potential, $\beta\mu$. The volume of the simulation cell was fixed during a simulation run. The chemical-potential reference state was that of an ideal gas with no interactions, so that at low densities the relationship

$$\beta\mu \rightarrow \ln(\rho\sigma^3) \quad \text{for } \rho \rightarrow 0 \quad (1)$$

holds, with $\rho=\langle N \rangle/V$, where N is the number of spheres present in the simulation box.

Results from multiple runs with overlapping particle number histograms were combined using the approach of Ferrenberg and Swendsen.^{25,26} Pressure was determined from the relationship $\ln \Xi = \beta P V$, where Ξ is the grand partition function. While the absolute value of Ξ cannot be obtained, the ratio of the partition function for two overlapping runs, and thus the difference in pressure, can be determined. The absolute value of the pressure for isotherms that contain no phase transitions can be obtained by including simulations at sufficiently low values of the chemical potential so that the ideal-gas equation of state is followed. For the high-density

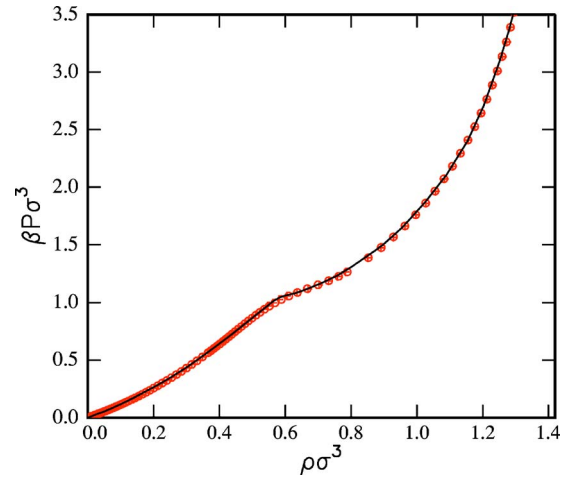


FIG. 2. (Color online) Pressure vs density for $\sigma=\sqrt{2}$. Points are from simulations with $L=8$ (circles) and $L=12$ (crosses). Solid line is from Gaunt (Ref. 17).

portions of isotherms that contain strong phase transitions, we were able to match results for systems with coarser discretization to an excellent accuracy. This is because the ordered solids that form have identical structures and the equation of state is dominated by vacancy effects that are independent of the discretization for this range of values. In particular, the pressures for the $\sigma=2\sqrt{2}$ and $\sigma=3\sqrt{2}$ systems were matched to those of $\sigma=\sqrt{2}$ and those for the $2\sqrt{3}$ system were matched to the $\sqrt{3}$ system. The $\sqrt{3}$ system contains only a weak transition and thus can be directly linked to low-density states.

III. RESULTS AND DISCUSSION

Table I summarizes the models studied and the results for the location of the phase transition. Numerical data for the pressure and density as functions of the chemical potential are available as supplementary material to this article.²⁷ In all cases multiple system sizes were studied and the results plotted against $1/L$ and extrapolated to infinite system size. Uncertainties listed in the table are due primarily to the extrapolation to infinite system size ($1/L \rightarrow 0$), rather than simulation statistical uncertainties, which were typically less than 0.1% relative error in density at a given chemical potential and system size.

Pressure-volume data obtained for $\sigma=\sqrt{2}$ are shown in Fig. 2, along with the series expansion results of Gaunt.^{17,28}

TABLE I. Excluded sites and location of fluid-solid transitions for hard-sphere models. Results have been extrapolated to $L \rightarrow \infty$.

σ	M	$\rho_f \sigma^3$	$\rho_s \sigma^3$	$\beta P \sigma^3$	$\beta \mu$
$\sqrt{2}$	7	0.59±0.01 (second-order transition)		1.05±0.02	1.09±0.01
$2\sqrt{2}$	81	0.696±0.002	1.237±0.001	3.10±0.01	5.19±0.01
$3\sqrt{2}$	305	0.778±0.002	1.355±0.002	4.62±0.02	7.56±0.02
$\sqrt{3}$	19	0.54±0.01	0.67±0.01	1.32±0.01	2.18±0.01
$2\sqrt{3}$	171	0.708±0.002	1.224±0.002	3.79±0.02	6.59±0.02
2	27	no transition			
3	93	0.88±0.02	1.00±0.02	4.5±0.1	6.7±0.2

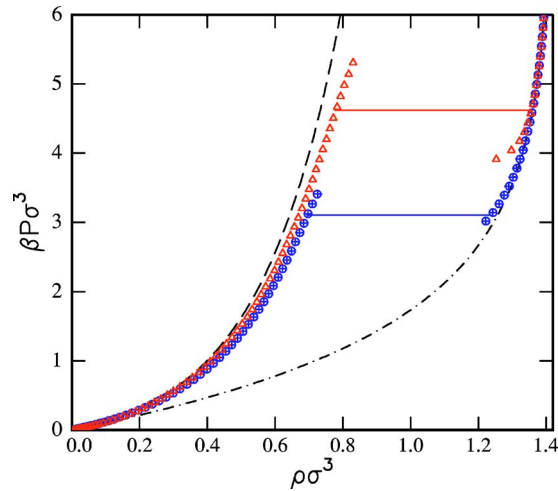


FIG. 3. (Color online) Pressure vs density for $\sigma=2\sqrt{2}$ (circles, $L=24$ and crosses, $L=16$) and $\sigma=3\sqrt{2}$ (triangles, $L=24$). Horizontal solid lines correspond to first-order phase transitions determined from the simulation data as explained in the text. Dashed line is the Carnahan-Starling equation for the continuum fluid and dash-dotted line the limiting behavior for the solid from a high-density expansion as explained in the text.

The isothermal compressibility was obtained as a function of system size between $L=8$ and $L=32$ and analyzed to obtain the approximate location of the transition in order to establish the validity of our code. Detailed finite-size scaling studies of the transition are already available for this system.^{19,29} Density, pressure, and chemical potential at the transition agree well with prior results. Gaunt¹⁷ gives $\beta\mu=1.12\pm0.07$, $\beta P\sigma^3=1.07\pm0.03$, and $\rho\sigma^3=0.60\pm0.03$ while Herringa and Blöte¹⁹ give $\beta\mu=1.094$ and Yamagata²⁹ $\beta\mu=1.097$. The last two studies do not report the transition density.

The next two models studied were those with $\sigma=2\sqrt{2}$ and $3\sqrt{2}$. The maximum packing density for both models is the same as for the continuum model, $\rho_{\max}\sigma^3=\sqrt{2}$, and corresponds to a fcc arrangement of the particles—this is also the case for the $\sigma=\sqrt{2}$ model. All these models are coarse enough that when the density is slightly reduced from the maximum packing density particles cannot move from their position in the crystal, being “pinned” by the underlying lattice even if there are nearby vacancies. A density expansion in this case gives that the pressure at both high and low densities must be

$$\beta P \rightarrow -\rho_{\max} \ln\left(1 - \frac{\rho}{\rho_{\max}}\right) \quad \text{for } \rho \rightarrow \rho_{\max} \text{ and } \rho \rightarrow 0. \quad (2)$$

Figure 3 shows the calculated isotherms along with the approximate Carnahan-Starling equation of state for the fluid³⁰ and the result of the density expansion of Eq. (2). While the Carnahan-Starling equation of state fails at high densities and is not the most accurate available for the fluid state (e.g., see Ref. 31), it is sufficiently accurate for comparison purposes in the present work. Unlike the $\sigma=\sqrt{2}$ system, for both $\sigma=2\sqrt{2}$ and $3\sqrt{2}$ systems the simulations show a first-order fluid-solid transition. Because of the transition, there is strong hysteresis of the density versus the chemical-potential curves, depending on whether the initial conditions

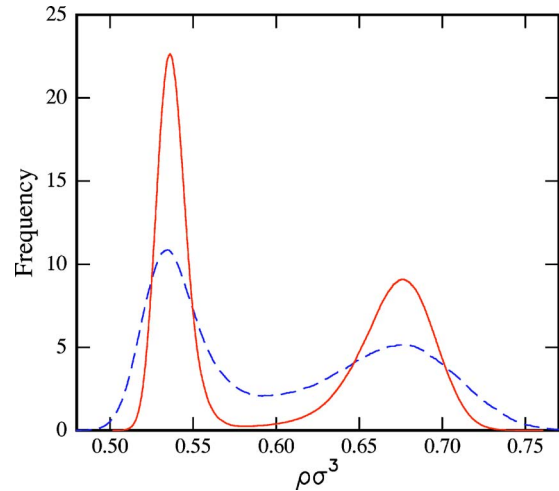


FIG. 4. (Color online) System size dependence of the density histograms at fluid-solid coexistence for $\sigma=\sqrt{3}$. Solid line is for $L=28$, $\beta\mu=2.179$ and dashed line for $L=18$, $\beta\mu=2.174$.

are in the low-density (disordered) or high-density (ordered) phase. The location of the transition was determined from the intersection of the chemical potential—pressure curves for the fluid and solid branches of the isotherms.

The density increases upon crystallization by 78% and 74% of the liquid density, respectively, for the $2\sqrt{2}$ and $3\sqrt{2}$ systems. These values are much greater than the corresponding modest increase in density for the continuum fluid⁵ ($\rho_f\sigma^3=0.943\pm0.004$, $\rho_s\sigma^3=1.041\pm0.004$, and $\beta P\sigma^3=11.7\pm0.2$). The transition pressure and chemical potential increase with increasing discretization, as expected.

The next two models studied were $\sigma=\sqrt{3}$ and $\sigma=2\sqrt{3}$. The $\sigma=\sqrt{3}$ model has ‘next-nearest-neighbor exclusion’ and forms a bcc solid with maximum density $\rho_{\max}=\frac{3}{4}\sqrt{3}$. The transition for $\sigma=\sqrt{3}$ is relatively weak, so that direct coexistence between disordered and ordered states can be observed within a single simulation run, as seen in Fig. 4. For this system, the transition chemical potential was determined from the condition of equality of areas for the fluid and solid histogram peaks. Calculations were performed for system sizes from $L=18$ to $L=28$. For smaller systems there is significant overlap between the two peaks, which makes an unambiguous determination of the two areas difficult. For larger systems, the free-energy barrier for interconversion between phases increases. Results for the coexistence chemical potential and fluid and solid densities were found to be approximately independent of the system size over the range of system sizes studied. Pressure-density isotherms for these two systems are plotted in Fig. 5. As seen for the $\sqrt{2}$ family of systems, the isotherms follow the continuum system equation of state increasingly well as the discretization is increased; the high-density isotherms approach closely the limiting values of Eq. (2). The transition pressure and chemical potential increase as the discretization is increased and are listed in Table I.

The last class of systems studied is for particle diameter an integer multiple of the lattice spacing (Fig. 6). For $\sigma=1$, Eq. (2) is exact, with $\rho_{\max}\sigma^3=1$. The $\sigma=2$ system also has $\rho_{\max}\sigma^3=1$ and the ground state is disordered as particles can

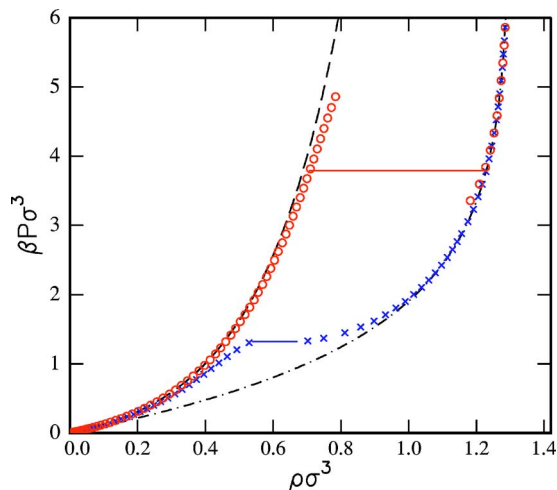


FIG. 5. (Color online) Pressure vs density for $\sigma=\sqrt{3}$ (x's, $L=24$) and $\sigma=2\sqrt{3}$ (circles, $L=24$). Lines are as for Fig. 3.

slide by half a particle diameter with respect to the surrounding particles while preserving full packing. The maximum density and particle arrangement for $\sigma=3$ have not been determined. The high-density phase appears disordered, irrespective of the simulation box size. The phase transition, however, is clearly (but weakly) first order, as seen previously for $\sigma=\sqrt{3}$.

The densities versus the imposed chemical potential are shown in Fig. 7 for a representative of each of the three classes of systems studied. Results from the simulations of the discretized models are compared to the exact values for $\sigma=1$, $\rho=1/[1+\exp(-\beta\mu)]$ and to the values from analytical integration of the Carnahan-Starling equation of state,³⁰ as reported in Ref. 32. As seen previously for the pressure-density isotherms, the discretized systems follow the continuum fluid values quite closely in the case of the systems with σ an integer multiple of $\sqrt{2}$ or $\sqrt{3}$ times the lattice spacing, but significantly less so in the systems for which σ is an integer multiple of the lattice spacing.

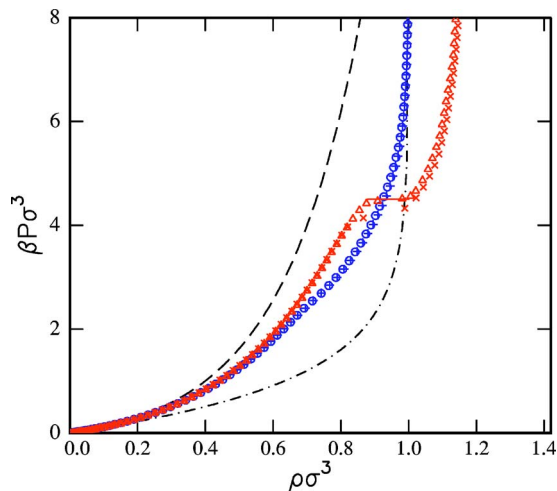


FIG. 6. (Color online) Pressure vs density for $\sigma=2$ (crosses, $L=18$; circles, $L=24$) and $\sigma=3$ (x's, $L=18$; triangles, $L=24$). Horizontal solid line corresponds to first-order phase transition for $\sigma=3$ determined from the simulation data as explained in the text. Dashed line is the Carnahan-Starling equation for the continuum fluid and dash-dotted line the exact expression for $\sigma=1$.

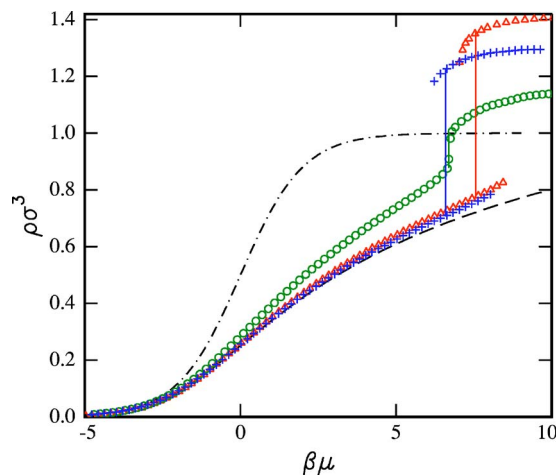


FIG. 7. (Color online) Density vs chemical potential for $\sigma=2\sqrt{3}$ (crosses), $\sigma=3$ (circles), and $\sigma=3\sqrt{2}$ (triangles). All systems are for $L=24$. Vertical solid lines correspond to first-order phase transitions determined from the simulation data as explained in the text. Dashed line is from the Carnahan-Starling equation for the continuum fluid and dash-dotted line the exact expression for $\sigma=1$.

In conclusion, it has been determined that the fluid thermodynamic properties of finely discretized hard-sphere systems converge relatively quickly but properties of the solid phases do not. Large lattice artifacts are seen in the location and character of order-disorder phase transitions. Integer values of the discretization parameter ζ result in significantly worse agreement with data for the continuum fluid than comparable values that are multiples of the nearest-neighbor and next-nearest neighbor distances. Numerical data for the chemical-potential-density-pressure relationships are available from the present study²⁷ that can be used to test the accuracy of theoretical approaches.

ACKNOWLEDGMENTS

The author gratefully acknowledges financial support for this work by the Department of Energy, Office of Basic Energy Sciences (Grant No. DE-FG201ER15121) and additional support by ACS-PRF (Grant No. 38165-AC9).

- ¹S. Auer and D. Frenkel, J. Chem. Phys. **120**, 3015 (2004).
- ²G.-W. Wu and R. J. Sadus, AIChE J. **51**, 309 (2005).
- ³X.-Z. Wang, J. Chem. Phys. **122**, 044515 (2005).
- ⁴G. Bryant, S. R. Williams, L. Qian, I. K. Snook, E. Perez, and F. Pincet, Phys. Rev. E **66**, 060501 (2002).
- ⁵W. G. Hoover and F. H. Ree, J. Chem. Phys. **49**, 3609 (1968).
- ⁶A. Z. Panagiotopoulos and S. K. Kumar, Phys. Rev. Lett. **83**, 2981 (1999).
- ⁷A. Ciach and G. Stell, Phys. Rev. Lett. **91**, 060601 (2003).
- ⁸A. Ciach and G. Stell, Phys. Rev. E **70**, 016114 (2004).
- ⁹S. Moghaddam, Y. C. Kim, and M. E. Fisher, J. Phys. Chem. B **109**, 6824 (2005).
- ¹⁰Q. Yan and J. J. de Pablo, Phys. Rev. Lett. **91**, 018301 (2003).
- ¹¹D. W. Cheong and A. Z. Panagiotopoulos, J. Chem. Phys. **119**, 8526 (2003).
- ¹²Y. C. Kim and M. E. Fisher, Phys. Rev. Lett. **92**, 185703 (2004).
- ¹³A. Z. Panagiotopoulos, J. Chem. Phys. **112**, 7132 (2000).
- ¹⁴D. M. Zuckerman, J. Phys. Chem. B **108**, 5127 (2004).
- ¹⁵S. Natarajan and J. K. Maranas, J. Chem. Phys. **118**, 9053 (2003).
- ¹⁶P. A. Monson and D. A. Kofke, Adv. Chem. Phys. **115**, 113 (2000).
- ¹⁷D. S. Gaunt, J. Chem. Phys. **46**, 3237 (1967).

- ¹⁸L. Lafuente and J. A. Cuesta, Phys. Rev. E **68**, 066120 (2003).
- ¹⁹J. R. Heringa and H. W. J. Blöte, Physica A **232**, 369 (1996).
- ²⁰C. K. Hall and G. Stell, Phys. Rev. A **7**, 1679 (1973).
- ²¹E. R. Cowley, J. Chem. Phys. **71**, 458 (1979).
- ²²L. Lafuente and J. A. Cuesta, J. Chem. Phys. **119**, 10832 (2003).
- ²³L. Lafuente and J. A. Cuesta, Phys. Rev. Lett. **93**, 130603 (2004).
- ²⁴A. Z. Panagiotopoulos, J. Phys.: Condens. Matter **12**, R25–R52 (2000).
- ²⁵A. M. Ferrenberg and R. H. Swendsen, Phys. Rev. Lett. **61**, 2635 (1988).
- ²⁶A. M. Ferrenberg and R. H. Swendsen, Phys. Rev. Lett. **63**, 1195 (1989).
- ²⁷See EPAPS Document No. E-JCPSA6-123-512530 for numerical data. This document can be reached via a direct link in the online article's HTML reference section or via the EPAPS homepage (<http://www.aip.org/pubservs/epaps.html>).
- ²⁸The contents of Figs. 12 and 13 of Ref. 17 seem to have been reversed—the data plotted here are from Fig. 13, labeled incorrectly in the original paper as “isotherm for body-centered cubic lattice gas.” This reversal is confirmed by the values of the transition grand potentials and densities given in Eqs. (4.36) and (4.38) of Ref. 17.
- ²⁹A. Yamagata, Physica A **215**, 511 (1994).
- ³⁰N. F. Carnahan and K. E. Starling, J. Chem. Phys. **51**, 635 (1969).
- ³¹M. Miandehy and H. Modarress, J. Chem. Phys. **119**, 2716 (2003).
- ³²P. Attard, J. Chem. Phys. **98**, 2225 (1993).